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Manual intraosseous device and method for introducing a bone portal to a predetermined depth in a bone

Abstract

A manual intraosseous (IO) device for introducing a bone portal to a predetermined depth relative to a surface of a target bone, and methods for using same. The device is manually actuated and configured to limit the depth of penetration of a bone portal into a bone cortex independently of an applied driving force and independently of subcutaneous tissue thickness surrounding the bone cortex, while providing terminal feedback that the bone portal has been penetrated to the predetermined depth. The device has transfusion facilitating components that remain fixed at the penetration site following penetration of the bone portal to the predetermined depth and releasable components that are separated from the transfusion facilitating components prior to a transfusion operation. The device may also include a stabilizer, or base, that allows a health practitioner to easily locate the insertion site and allows low profile fixation of the IO transfusion component (IOcatheter).

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Background/Summary

RELATED APPLICATIONS (1) This is a continuation application of U.S. patent application Ser. No. 18/280,767, filed on Sep. 7, 2023, now U.S. Pat. No. 12,004,776, issued on Jun. 11, 2024, which is a 371 National Stage Entry of PCT/IB22/55857, filed on Jun. 24, 2022, which claims priority to Israeli Patent Application No. 284380, filed Jun. 24, 2021, the disclosures of which are incorporated herein by reference in their entirety.

FIELD OF THE INVENTION

(1) The present invention relates to the field of intraosseous devices. More particularly, the invention relates to a manual intraosseous device and method for introducing a bone portal to a predetermined depth relative to a bone surface in order to reliably and safely access the bone marrow cavity.

BACKGROUND OF THE INVENTION

(2) The administration of medication to an injured or critically ill patient is many times delayed due to the difficulty in establishing an intravenous line. During such situations, a lifesaving alternative by which vascular access is quickly achieved is through intraosseous (IO) infusion, whereby fluids and medications are injected into a marrow cavity of a long bone such as the femur, tibia and humerus, or of the sternal bone in the manubrium. Fluids drain into a central venous canal and is then carried to the bloodstream.

(3) The success of an IO infusion procedure is contingent upon penetration of the bone cortex to a patient-specific depth in order to access the bone marrow. The bone marrow will not be accessible if tissue overlying a target bone is not sufficiently penetrated, for example when an incorrect needle length is employed or an excess amount of subcutaneous tissue exists, or alternatively if the needle is deployed to an excessive depth, resulting in possible damage to healthy surrounding tissues and organs when the bone is overpenetrated after the needle has penetrated two opposite diametric regions of the bone.

(4) Particularly, the bones of infants are very thin and are sometimes concealed by excessive overlying soft tissue. A health practitioner performing an IO penetration procedure therefore requires a high level of accuracy to determine the proper depth of penetration for the IO needle.

(5) Penetration of the sternum, for example, presents a high risk in overpenetration of its manubrium, which is joined to the clavicles and the cartilages of the first pair of ribs, due to the relatively thin marrow cavity of the sternum. A needle that unintentionally penetrates the distal cortex of the manubrium is liable to injure vital body parts such as the heart, lungs and the great vessels associated with the heart.

(6) Bones having a thicker wall than the sternum, such as long bones in adults, are also subject to overpenetration if the health practitioner applies an excessive penetration force, which may continue to be applied even after the needle has accessed the bone cavity.

(7) It would be desirable to provide an IO device for preventing deeper, potentially harmful penetration after the health practitioner has introduced the bone portal to a predetermined depth within a bone cortex.

(8) Some prior art IO devices are known to limit penetration; however, they do not provide an indication to the health practitioner that the bone portal has been introduced to the predetermined depth, and consequently the health practitioner often continues to apply a penetrating force. In addition to the unnecessary effort that is being exerted, the continuously or excessively applied penetrating force transmitted to the intraosseous needle caused to be stationary is liable to injure surrounding body structures and even lead to a bone fracture in the vicinity of the penetration site.

SUMMARY OF THE INVENTION

(9) One aspect of the invention is a manual intraosseous device for introducing a bone portal to a predetermined depth relative to a surface of a target bone, comprising an intraosseous catheter configured with a bone portal that is penetrable into a cortex of the target bone; a releasable component assembly that is releasably attached to the intraosseous catheter, by which a manual axial force is transmittable to the intraosseous catheter to initiate corresponding axial penetration into subcutaneous tissue associated with the target bone as well as into the cortex of the target bone; and a penetration depth limiting mechanism comprising at least first and second elements provided with the component assembly that are separate from each other when the bone portal is penetrated within the bone cortex by less than a predetermined depth relative to an outer surface of

the bone cortex which is independent of a thickness of the subcutaneous tissue and are secured to each other when the bone portal is penetrated within the bone cortex to the predetermined depth, to prevent any additional distal displacement of the bone portal. The penetration depth limiting mechanism is a terminal feedback indicating mechanism which is configured to provide a tactile indication that the intraosseous catheter is either proximally or distally displaceable following application of the manual axial force in a corresponding direction when the bone portal is penetrated within the bone cortex by less than the predetermined depth and that the intraosseous catheter is prevented from undergoing proximal or distal displacement after the bone portal has been penetrated within the bone cortex to the predetermined depth despite application of the manual axial force. The intraosseous catheter is further configured with an internal lumen through which an infusion liquid or a bone marrow aspirate is flowable, and at least one port by which the internal lumen is positionable in fluid communication with a bone marrow cavity of the target bone, and is connectable with a connector interface that induces flow of the infusion liquid or the bone marrow aspirate when the component assembly is released from the intraosseous catheter.

(10) Another aspect of the invention is a manual intraosseous device for introducing a bone portal to a predetermined depth relative to a surface of a target bone, comprising an intraosseous catheter configured with a driving member and a bone portal that is penetrable into a cortex of the target bone in conjunction with the driving member; a driving member hub configured to carry the intraosseous catheter; an inner tubular sleeve; an outer tubular sleeve concentric to, and axially displaceable relative to, the inner sleeve; detachable connection means for releasably connecting the intraosseous catheter hub from the outer sleeve member, by which a manual axial force applied to the driving member hub is transmittable to the intraosseous catheter to initiate corresponding axial penetration into subcutaneous tissue associated with the target bone as well as into the cortex of the target bone; and a penetration depth limiting mechanism comprising at least first and second elements provided with the inner sleeve and outer sleeve, respectively, which are separate from each other when the bone portal is penetrated within the bone cortex by less than a predetermined depth relative to an outer surface of the bone cortex which is independent of a thickness of the subcutaneous tissue and are secured to each other when the bone portal is penetrated within the bone cortex to the predetermined depth, to prevent any additional distal displacement of the bone portal. The intraosseous catheter is further configured with an internal lumen through which an infusion liquid or a bone marrow aspirate is flowable, and at least one port by which the internal lumen is positionable in fluid communication with a bone marrow cavity of the target bone, and is connectable with a connector interface that induces flow of the infusion liquid or the bone marrow aspirate when the driving member hub is released from the intraosseous catheter.

(11) Another aspect of the invention is a stabilizer for an intraosseous device, comprising a base configured to be affixed to a skin surface and to be detachably coupled with a structure of the intraosseous device, wherein the base is bored with an aperture that is adapted to overlie a penetration site and receive a needle assembly of the intraosseous device; and a surface surrounding the base which is configured with a plurality of guidable peripheral edges. Each of the guidable edges has a distinctive shape that facilitates positioning close to a prominent anatomical feature and that is located at a predefined age-specific or site-specific distance from the aperture.

(12) Another aspect of the invention is a combination of a stabilizer for an intraosseous device and a flexible transfusion tube, comprising a base configured to be affixed to a skin surface and to be detachably coupled with a structure of the intraosseous device, wherein the base is bored with an aperture that is adapted to overlie a penetration site and receive a needle assembly of the intraosseous device for use during a penetration operation, and is additionally formed with a plurality of differently oriented notches; and a flexible tube removably connected to the needle assembly and within a lumen of the flexible tube a solid driving member is insertable during the penetration operation, wherein the flexible tube, following removal of the driving member therefrom, is in fluid communication with a connector interface that induces flow of infusion liquid

or bone marrow aspirate during a transfusion operation. The flexible tube is securable in one of the notches during the transfusion operation to facilitate secured connection to the base interface.

(13) Another aspect of the invention is a method for accessing a bone marrow cavity, comprising the steps of pinpointing a prominent anatomical landmark associated with a suitable penetration site; aligning a guidable edge of a stabilizing base configured with a predetermined geometrical relation between the guidable edge and an aperture bored in the base which is adapted to overlie the penetration site, with the prominent anatomical landmark; applying a manual axial force by an intraosseous device body that causes an intraosseous catheter attached therewith to be introduced through the aperture and to penetrate a cortex of a target bone, until the intraosseous catheter has been penetrated to a predetermined depth relative to a surface of the target bone that ensures access of the intraosseous catheter to a bone marrow cavity of the target bone, wherein additional distal penetration of the intraosseous catheter is prevented despite additional application of distal manual axial force by the intraosseous device after the intraosseous catheter has been penetrated to the predetermined depth; and detaching the intraosseous device body from the intraosseous catheter and base, fixation of the intraosseous catheter to built-in notch in the base to allow a low profile fixation of the tube and a secure fluid transport from and to the bone

(14) Yet another aspect of the invention is a manual intraosseous device for introducing a bone portal to a predetermined depth relative to a surface of a target bone, comprising: an intraosseous needle assembly including an intraosseous catheter configured to facilitate transfusion of fluids and aspiration of bone marrow and having a bone portal adapted to penetrate into subcutaneous tissue adjacent the target bone and the cortex of the target bone and a flexible tube through which liquids can be conveyed to and from the bone marrow of the target bone, and a driving member releasably attached to the intraosseous catheter and by which a manual axial force is transmittable to the intraosseous catheter to initiate axial penetration into the subcutaneous tissue and cortex of the target bone; an inner tubular sleeve arranged concentric to the intraosseous catheter; an outer sleeve member including an outer sleeve arranged concentric to the inner sleeve and being axially displaceable relative to an outer surface of the inner sleeve and a grip engaging an outer surface of outer sleeve, the outer sleeve member operably connected to the driving member, and wherein the driving member includes a rod and a driving member hub engaging the outer sleeve member and having a throat portion receiving a proximal end of the rod therein; a probe needle assembly engaged with the outer sleeve member and at least partially received within an interior of the inner sleeve, and including at least one bone cortex-contacting probe needle, a motion inducer and an annular needle holder securing the at least one probe needle therein; a proximal spring positioned within the outer sleeve member, operably connected to the probe needle assembly, and configured to limit the depth that bone portal penetrates the target bone; and a distal spring positioned within the inner sleeve, operably connected to the probe needle assembly, and configured to move the outer sleeve member to a starting position if the bone portal penetrates the target bone cortex to the predetermined depth, to prevent any additional distal displacement of the bone portal. The proximal and distal springs provide a terminal feedback indicating mechanism configured to provide a tactile indication that the intraosseous catheter is either proximally or distally displaceable following application of the manual axial force in a corresponding direction when the bone portal penetrates the bone cortex by less than the predetermined depth and that the intraosseous catheter is prevented from undergoing proximal or distal displacement after the bone portal has penetrated the bone cortex to the predetermined depth despite application of the manual axial force.

(15) Still another aspect of the invention is a manual intraosseous device for introducing a bone portal to a predetermined depth relative to a surface of a target bone, comprising: an intraosseous catheter configured with a driving member and a bone portal that is penetrable into a cortex of the target bone in conjunction with the driving member; a driving member including a driving member hub configured to support the intraosseous catheter; an inner tubular sleeve; an outer tubular sleeve concentric to, and axially displaceable relative to, the inner sleeve; wherein the driving member

hub is connected to the outer sleeve, by which a manual axial force applied to the driving member hub is transmittable to the intraosseous catheter to initiate corresponding axial penetration into subcutaneous tissue associated with the target bone and into the cortex of the target bone; and a penetration depth limiting mechanism comprising at least first and second elements provided within the inner sleeve and outer sleeve, respectively, which are separate from each other when the bone portal is penetrated within the bone cortex by less than a predetermined depth relative to an outer surface of the bone cortex which is independent of a thickness of the subcutaneous tissue and are secured to each other when the bone portal is penetrated within the bone cortex to the predetermined depth, to prevent any additional distal displacement of the bone portal.

(16) It is an object of the present invention to provide a manual intraosseous device that reliably limits the depth relative to a bone surface to which a bone portal is introduced regardless of the magnitude of force that is applied and of the thickness of subcutaneous tissue.

(17) It is an object of the present invention to provide a manual intraosseous device for introducing a bone portal to a predetermined depth relative to a bone surface that can be withdrawn in its entirety from the tissue if the bone cortex has not been penetrated at all or if the bone portal has not been introduced to the predetermined depth within the marrow cavity.

(18) It is an additional object of the present invention to provide a manual intraosseous device capable of generating a reliable indication that the bone portal has been introduced to the predetermined depth and that the health practitioner should stop applying the penetrating force.

(19) Other objects and advantages of the invention will become apparent as the description proceeds.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Embodiments of the invention are further described but are in no way limited by the following illustrations.

(2) FIG. 1 is a top perspective view of an assembled manual IO device according to a first embodiment of the present invention;

(3) FIG. 1A is a cross-sectional view of components of the device of FIG. 1;

(4) FIG. 1B is a cross-sectional view of components of a manual IO device according to a second embodiment of the present invention;

(5) FIG. 1C is another cross-sectional view of components of the device of FIG. 1B;

(6) FIG. 2 is an exploded view of the device of FIGS. 1 and 1A;

(7) FIG. 2A is an exploded view of the device of FIG. 1B;

(8) FIG. 3 is top perspective view of a stabilizer and related elements according to an embodiment of the present invention for use with the manual IO devices of the present invention;

(9) FIG. 3A is top perspective view of a stabilizer and related elements according to another embodiment of the present invention for use with the manual IO devices of the present invention;

(10) FIG. 4 is detailed top perspective view of the stabilizer of FIG. 3;

(11) FIG. 4A is detailed top perspective view of the stabilizer of FIG. 3A;

(12) FIG. 5 is a cross-sectional view of an inner tube of the device of FIG. 1;

(13) FIG. 5A is a top perspective view of an inner tube of the device of FIG. 1B;

(14) FIG. 6 is a cross-sectional view of an outer sleeve member of the device of FIG. 1;

(15) FIG. 6A is a cross-sectional view of an outer sleeve member of the device of FIG. 1B;

(16) FIG. 6B is another cross-sectional view of the outer sleeve member of the device of FIG. 1B;

(17) FIG. 7 is a top perspective view of components of the device of FIG. 1;

(18) FIG. 8 is cross-sectional view of the components of FIG. 7;

(19) FIG. 9 is a cross-sectional view of a bone portal component of the device of FIGS. 1 and/or

1B;

(20) FIG. 9A is another cross-sectional view of a bone portal component of the device of FIGS. 1 and/or 1B;

(21) FIG. 10 is a pre-penetration cross-sectional view of the device of FIG. 1, as placed on a patient's skin for penetration into the patient's sternal bone cortex;

(22) FIG. 11 is a cross-sectional view of the device of FIG. 1 after a driving force has been applied to a driving member hub that causes outer sleeve member to be distally displaced with respect to inner sleeve of the device;

(23) FIG. 11A is a detailed view of FIG. 11;

(24) FIG. 12 is a cross-sectional view of the device of FIG. 1 after the outer sleeve member has been additionally displaced distally, until the distal end of probe needles of the device contact a bone cortex.

(25) FIG. 12A is a detailed view of FIG. 12;

(26) FIG. 13 is a cross-sectional view of the device of FIG. 1 after the manual driving force continues to be applied onto driving member hub and/or outer sleeve member of the device of FIG. 1 following the immobilization of the probe needles;

(27) FIG. 13A is a first detailed view of FIG. 13 (top);

(28) FIG. 13B is a second detailed view of FIG. 13 (bottom);

(29) FIG. 14 is a cross-sectional view of the device of FIG. 1 with continued application of the manual driving force;

(30) FIG. 14A is a first detailed view of FIG. 14 (top);

(31) FIG. 14B is a second detailed view of FIG. 14 (bottom);

(32) FIG. 15A is a cross-sectional view of the device of FIG. 1 with the distal end of IO needle assembly deployed to a predetermined and maximum depth within the patient's sternal bone cortex;

(33) FIG. 15B is a cross-sectional view of the IO catheter penetrating the patient's sternal bone cortex with the releasable component assembly of the device of FIG. 1 having been pulled and proximally displaced away therefrom;

(34) FIG. 16 is a top perspective view of the stabilizer of FIG. 3 with a flexible tube of the IO catheter secured in place therein;

(35) FIG. 17 is a top perspective view of a device package for use with the devices of FIGS. 1 and 1B; and

(36) FIG. 18 is a plan view of an embodiment of a driving member hub for use with the devices of FIGS. 1 and 1B.

DETAILED DESCRIPTION OF THE INVENTION

(37) The manually actuated intraosseous (IO) device is configured to limit the depth of penetration of a bone portal into a bone cortex independently of an applied driving force and independently of subcutaneous tissue thickness surrounding the bone cortex, while providing terminal feedback that the bone portal has been penetrated to the predetermined depth. The IO device has transfusion facilitating components that remain fixed at the penetration site following penetration of the bone portal to the predetermined depth and releasable components that are separated from the transfusion facilitating components prior to a transfusion operation. In some embodiments, the device may also include a stabilizer, or base, that allows a health practitioner to easily locate the insertion site and allows low profile fixation of the IO transfusion component (IO catheter).

(38) As referred to herein, a "transfusion" operation is meant to include both infusion of liquids into the bone marrow cavity and aspiration of bone marrow from the bone marrow cavity.

(39) FIG. 1 illustrates an assembled manually actuated IO device 30 prior to penetration into a target bone, according to a first embodiment. The IO device 30 comprises an inner tubular sleeve 10, and an outer tubular sleeve 20 concentric to inner sleeve 10 and carrying a probe needle assembly arranged such that the outer sleeve 20 is axially displaceable relative to the outer tubular surface of inner sleeve 10, i.e., in a direction parallel to the axis of a main intraosseous needle

assembly adapted to penetrate into adjoining tissue and to facilitate transfusion of life-saving fluids and aspiration of bone marrow. A grip **16**, such as one configured with a cutout **17**, e.g., of an inverted U-shape, may be integrally formed with the outer surface of outer sleeve **20** to assist in gripping the handle while pushing it in the proximal direction to axially displace it relatively to inner sleeve **10**. The combination of grip **16** and outer sleeve **20** defines an outer sleeve member **15**. In other words, the outer sleeve member **15** constitutes a “handle” of the device **30** that includes the outer sleeve **20** and grip **16** (and in one embodiment also includes the inner part of outer sleeve **20**). The addition of grip **16** increases the lateral dimension of the outer sleeve member **15** so that the latter is able to assume a non-circular cross section, although a circular cross section is also within the scope of the invention.

(40) Various elements and embodiments of the IO device **30** will now be discussed.

(41) In one embodiment outer sleeve **20** includes a solid internal structure (also referred to as outer tubular sleeve **20**). In one such embodiment, outer sleeve **20** includes an inner structure to be assembled with inner sleeve **10**, as shown in FIG. **1A**. In another embodiment, outer sleeve **200** includes a separate inner part, referred to herein as a body **4**, that is assembled into outer sleeve **200** (i.e., outer sleeve **200** is assembled onto body **4**) as described hereinbelow and shown in FIGS. **1B** and **1C**.

(42) FIG. **1A** is an axial cross-sectional view showing one embodiment in which outer sleeve **20** is one solid part with an inner structure. To ensure that inner sleeve **10** and outer sleeve **20** remain in a mutual concentric relation during axial displacement, a protrusion **22**, e.g., with a pointed end, protruding radially outwardly from outer surface **11** of inner sleeve **10** is received in an axial slot **13** formed in the inner surface of outer sleeve **20**, as shown in FIG. **1A**. In some embodiments, two diametrically opposite slots **13** are provided, each configured to receive a corresponding protrusion **22** therein. Outer sleeve **20** may include a stopper **23** protruding inwardly from the inner surface of outer sleeve **20**, located distally to the distal end of slot **13**, to limit the proximal displacement of outer sleeve **20** relative to inner sleeve **10**. Both the distal edge of protrusion **22** and the proximal edge of stopper **23** may be planar to improve the engagement therebetween.

(43) FIG. **1B** illustrates another embodiment in which outer sleeve **200** is assembled onto body **4** to form outer sleeve member **15**. The engagement between inner sleeve **100** and outer sleeve member **15** is carried out by means of additional elements as will be described hereafter and shown in FIGS. **1B** and **1C**.

(44) Body **4** includes an annular mounting post **51** protruding outwardly from an outer surface of body **4**. A motion inducer **28** is inserted through annular mounting post **51** of body **4** while snaps **77** of motion inducer **28** (see FIG. **7**) aligned to slots in the body **4** and reach its final position on the body **4** when slots become an open tunnel (i.e., are in fully open positions). In some embodiments, this is an undetachable connection.

(45) In some embodiments, an annular needle holder **26** is also inserted through annular mounting post **51**, and these two elements have a similar diameter (ID and OD, respectively) so that there is a friction between the two elements that holds them connected (i.e., an interference or friction fit). To strengthen the connection between these two parts, further connecting means may be included in some embodiments (i.e., in addition to a friction fit). Non-limiting examples of further connecting means may include snaps or pins that are received in corresponding channels or apertures, or a c-clips that prevent (distal) movement of annular needle holder **26**.

(46) In the embodiment shown in FIGS. **1B** and **1C**, c-clip **223** is inserted on to annular mounting post **51** of body **4** and to the final position inside slot **224**, and thereby prevents detachment of inner sleeve **100** from body **4**.

(47) To ensure that inner sleeve **100** and body **4** remain in a mutual concentric relation during axial displacement, a protrusion **222**, e.g., with a pointed end, protruding radially inwardly from outer surface **111** of inner sleeve **100** is received in an axial slot **6** formed between the outer surface of body **4** and the inner surface of outer sleeve **200**. In some embodiments, two or more ribs **113**

protrude outwardly from outer surface **111** of inner sleeve **100** are received in axial slots formed in the inner surface of outer sleeve **200**.

(48) Annular needle holder **26** is inserted through mounting post **51** protruding in outwardly from the inner surface of body **4** to limit the proximal displacement of body **4** relative to inner sleeve **100**. To improve the engagement therebetween, both the distal edge of protrusion **222** and the proximal surface of needle holder **26** may be planar.

(49) Outer sleeve member **200** is mounted onto body **4** and thereby that prevents outwardly movement of protrusion **222**. Outer sleeve member **200** and body **4** are connected in an undetachable connection by means of screws, snaps, or any other similar means.

(50) Referring again to FIGS. **1** and **2**, a driving member hub **5** configured to support (i.e., carry) an IO needle assembly **35** is provided in an abutting relation with a proximal surface of the outer sleeve member **15**. Driving member hub **5** may be configured with a domed proximal surface and with the same non-circular horizontal cross section as the outer sleeve member **15** and aligned therewith to improve a user's grip thereof. The distal end of inner sleeve **10** in turn is coupled with a stabilizer **40** through which the probe needles and tube assembly pass when penetrating the patient's underlying tissue.

(51) Outer sleeve member **15** is distally displaceable relative to inner sleeve **10** or **100** when a manual force is applied to driving member hub **5**, but is prevented from being displaced due to unintentional operation of IO device **30** when a safety latch **45** is coupled between outer sleeve **20** and inner sleeve **10** or outer sleeve **200** and inner sleeve **100**. In various embodiments, safety latch **45** may be configured with a finger-engageable ring and one or more coupling elements that are connected to the ring and pass through both inner sleeve **10**, **100** and outer sleeve **20**, **200** respectively, via aligned apertures, or may be configured in other ways well known to those skilled in the art.

(52) Safety latch **45** may be releasably coupled to inner sleeve **10**, **100** and outer sleeve **20**, **200**, or alternatively may be undetachably but movably connected to one of inner sleeve **10**, **100** or outer sleeve **20**, **200**.

(53) It will be appreciated that in alternate embodiments, IO device **30** may be provided without various components shown/described herein, e.g., the alignment protrusions, without a stabilizer and/or without a safety latch.

(54) Stabilizer **40** is adapted to be affixed to a skin surface during both a penetration operation and a transfusion operation. Inner sleeve **10** or **100** is adapted to be coupled to stabilizer **40** during a penetration operation and decoupled therefrom in response to a user action once the bone portal has been introduced to the predetermined depth within the bone. Stabilizer **40**, as shown in FIGS. **3/3A** and **4/4A**, is further discussed below.

(55) With reference to FIG. **2**, driving member hub **5** is shown to be separated from outer sleeve member **15**. Driving member hub **5** may be detachably coupled with outer sleeve member **15**, or alternatively may be undetachably connected to outer sleeve member **15** for one-time use embodiments of the IO device **30**. A proximal surface **19** of outer sleeve member **15** is adapted to be in abutting relation with an undersurface of driving member hub **5** that surrounds a cavity (i.e., a central opening) **18** provided within the outer sleeve member **15**. A wall **31** delimiting the cavity extends from the radial outward edge of an annular intermediate surface **33** of the outer sleeve member **15**, which is distally spaced from proximal surface **19**, to a terminal edge that is proximally spaced from proximal surface **19** (also shown in FIGS. **1A** and **1B**).

(56) Reference is made to FIGS. **7**, **8** and **10**, which show a probe needle assembly **25** engaged internally with an element of outer sleeve member **15** (or with an element of body **4**) and also partially received within the interior of inner sleeve **10**, **100**. Probe needle assembly **25** comprises an annular needle holder **26** to which one or more bone cortex-contacting probe needles **27** are secured via corresponding through-holes **24**. These probe needles **27** are inserted into bores at the distal end of motion inducer **28** in an undetachable connection e.g., via threading, adhesive or other

attachment means. The probe needles **27** pass through corresponding through-holes **24** of annular needle holder **26**. The shank of each probe needle **27** may have a uniform thickness, or alternatively may be configured with a radial protrusion. For example, in some embodiments the radial protrusion may protrude radially by a distance of at least 0.2 mm from the shank peripheral surface and may be spaced by a distance of up to 1 cm from the probe end. In various embodiments, a motion inducer **28** and/or a nut **29** are also provided, with nut **29** secured to annular mounting post **51** to prevent unwanted movement of the motion inducer **28** (see FIGS. **2** and **10**).

(57) With further reference to FIGS. **2**, **2A**, **7** and **8**, an intraosseous needle assembly **35** comprises an IO catheter **92** and a driving member. IO catheter **92** comprises a flexible tube **55** through which liquids can be conveyed to and from the bone marrow of a patient. The flexibility of tube **55** permits the tube to be folded after the penetration procedure is completed, to minimize tube protrusion from the patient's body. The proximal end of flexible tube **55** is connected to a female Luer-Lock fitting **57**, which is adapted to be connected to a standard connector.

(58) The driving member serves to transmit the manual force applied by the user to IO catheter **92** despite the flexibility of tube **55**. In various embodiments, the driving member constitutes a rod **42** with a blunt distal end **43**. Rod **42** is made of a metal (e.g., made of stainless steel). Rod **42** is received within the interior of tube **55**, and the outer diameter of the rod may be substantially equal to the inner diameter of tube **55**. In various embodiments, the proximal end of rod **42** is irremovably received, such as by frictional engagement, or removably received such as by threading, in an axial bore formed in a central throat portion **7** distally extending from driving member hub **5**. Throat portion **7** in turn is received in a central opening **18** provided within the intermediate surface **33** of outer sleeve member **15**, allowing the throat portion **7** to be securely engaged with the outer sleeve member **15**. The function of throat portion **7** is both to firmly fix the rod **42** in place and to ensure stability during penetration of IO catheter **92** through the patient's soft subcutaneous tissue and into the underlying bone.

(59) IO device **30** is resiliently displaceable by means of two compression springs **63** and **64** (see FIGS. **2** and **2A**). Proximal spring **63** assists in limiting the depth of penetration of bone portal **60**, as will be described hereinbelow. Distal spring **64** is instrumental in providing the terminal feedback by returning outer sleeve member **15** to a starting position if the bone portal **60** has not been deployed to a predetermined depth.

(60) Reference is made again to FIGS. **3**, **4** and **3A**, **4A**, which illustrate embodiments of stabilizer **40**. Stabilizer **40** includes a thin and planar supporting surface **37** (e.g., made from polycarbonate or other type of rigid plastic), from which proximally protrudes a base **38** adapted to be detachably coupled with the inner sleeve **10**. A thin fixation sticker **65** may be adhesively attachable to both a patient's skin surface adjoining the penetration site and to the distal side of supporting surface **37**. In various embodiments, adhesive is generally applied onto both sides of fixation sticker **65**, and a release liner (i.e., backing) **66** may be applied to the skin-facing side of fixation sticker **65** to protect the adhesive before use. In some embodiments, the adhesive is a pressure-sensitive adhesive.

(61) Fixation sticker **65** may have three arms **69a-c** extending in different directions from supporting surface **37** to increase the available surface area for connection of fixation sticker **65** to the patient's skin surface. The material of arms **69a-c** may also extend radially inwardly to underlie, and to be integral with, supporting surface **37**, and a foldable weakened area **67** may be provided between supporting surface **37** and a corresponding arm. In some embodiments, arms **69a-c** have an equal shape and/or size. In other embodiments, their shape and/or size of each of arms **69a-c** differ from each other.

(62) In various embodiments, base **38** has a circular shape with a circumferential edge **39** substantially perpendicular to supporting surface **37** and a slightly domed proximal surface **34**. In alternate embodiments, the base may have other shapes. In one embodiment, base **38** has a dome **34** with a solid/closed structure as shown in FIGS. **3** and **4**. In another embodiment, base **138** has a

dome **134** with a hollow/open structure as shown in FIGS. **3A** and **4A**, which reduces its weight and amount of material used to form same. Three notches **41a-c** are formed within base **38**, and radially extend from the center of base **38** to circumferential edge **39** to accommodate the passage therethrough of IO catheter **92**. In alternate embodiments, notches **41a-c** extend to circumferential edge **39** in another direction from a specific location on base **38**.

(63) One or more holes **46**, each of which is bored in proximal surface **34**, supporting surface **37** and in the underlying fixation sticker **65** surface and radially spaced from the center aperture of base **38**, accommodates the passage therethrough of a corresponding probe needle **27**.

(64) One or more of the notches **41a-c** may be configured with a curved supporting surface **49** proximally spaced from supporting surface **37** that curves distally from circumferential edge **39** to the center aperture of base **38**, for supporting flexible tube **55** when it is bent during a transfusion operation (see FIGS. **15B** and **16**). Circumferential edge **39** may be configured with two opposed constricting elements **32a** and **32b**, or any other number of constricting elements, e.g., triangularly shaped, at each notch that reduce the width of the given notch in order to prevent unintentional movement of the affixed tube. Elements **32a-b** also serve to couple with corresponding elements of the inner sleeve during a penetration operation.

(65) In some embodiments, the notches **41a-c** do not coincide with the center aperture of base **38**, as long as the flexible tube **55** is able to be suitably bent and supported.

(66) While the base **38** is shown having three notches **41a-c**, any other number of notches may be formed within base **38** in alternate embodiments, including, but not limited to, one, two, four, five, six, seven or eight notches.

(67) A portion **36** of base **38** adjoining the elongated back of each constricting element that does not protrude into a notch **41a, b** or **c** protrudes from circumferential edge **39**, to produce a socket, having for example a U-shape.

(68) Planar supporting surface **37** has a plurality of terminal edges **53** that are spaced from and facing the corresponding notch **41a, b** or **c**, and a plurality of guidable peripheral edges **54** extending between two adjacent terminal edges **53**. In various embodiments, terminal edges **53** can be straight or convex. In some embodiments a terminal edge may be circumferentially spaced from a notch, producing a socket, or unaligned therewith. In other embodiments supporting surface **37** may support a base provided without any notches. Each of the terminal edges **53** is shown to be aligned with a corresponding arm **69a, b** or **c** of fixation sticker **65**, extending across the width of the arm. Guidable edge **54** has a distinctive shape so as to be positionable close to a prominent anatomical feature (e.g., the sternal notch for a sternal application, a tibial tuberosity for a tibial application, or any other easy-to-find anatomical landmark). Guidable edge **54** is designed to be located at a predefined distance from the center of base **38**, through which IO catheter **92** passes, to assist in properly aligning stabilizer **40** relative to the penetration site. This is because supporting surface **37** and each of the guidable edges **54** is configured with a predetermined geometrical relation, which may be age-specific so as to be appropriate for a human body size of an average age and/or site-specific for a specific anatomical structure, between a guidable edge **54** and the aperture adapted to overlie the penetration site.

(69) Each of the guidable edges **54** is shown to be concave, making them suitable to be positioned adjacent to a prominent anatomical feature such as the sternal notch, but the guidable edges may be configured in other ways in alternate embodiments. All of the guidable edges **54** may be uniformly shaped. In alternate embodiments, each guidable edge may be differently shaped, such as with a different radius of curvature, or differently dimensioned, such as dimensioned with a different distance to the center of the base. By having similarly configured guidable edges **54**, the health practitioner performing the penetration operation may conveniently reposition the supporting surface **37** and the base **38** protruding therefrom, depending on the current angle of the health practitioner relative to the anatomical feature and on the configuration of the guidable edge **54**. By having differently configured guidable edges, the health practitioner performing the penetration

operation may easily position the IO device relative to different anatomical sites. For example, one guidable edge **54** may be configured to facilitate repositioning when it is positioned adjacent to the proximal tibia anatomical landmark location and another guidable edge **54** is positioned adjacent to the sternum anatomical landmark location.

(70) The three notches **41a-c** provided in base **38** advantageously allow the health practitioner to stand in one of three different directions relative to the patient, for example while standing behind the head or to the side of the patient, in order to initiate a transfusion operation with use of the IO catheter, thereby enhancing the convenience and ease of use.

(71) More importantly, the three notches **41a-c** provide increased flexibility in deciding how the flexible tube **55** of the IO catheter **92** should be connected to an infusion-related component. An important consideration is the location of the infusion tube within the congested environment of an ambulance or helicopter, where the available room surrounding the patient is limited and often crowded with paramedics or other emergency care assistants attempting to provide life-saving care. Since infusion bags are often hung above the patient in such a congested environment, the infusion tube often passes in the space above the patient as well. An emergency care assistant is therefore liable to unknowingly collide with the infusion tube and cause the corresponding needle to become dislodged from the patient's body. A stabilizer configured with spaced and differently positioned notches advantageously helps to secure the infusion tube in a position that is most protected and user friendly to the emergency care assistant while the patient is being transported.

(72) FIG. 5 illustrates a cross sectional view of inner sleeve **10**, prior to the outer sleeve member **15** being mounted thereabout. FIG. 5A illustrates the annular distal wall **113** of the inner sleeve **100**. Outer circumferential surface **11** of inner sleeve **10**, **100** may be formed with an aperture **16**, **116** within which safety latch **45** (see FIG. 1) is insertable. Inner surface **8** of inner sleeve **10** is formed with a plurality of axially spaced grooves **9**, each of which is recessed from inner surface **8**. In one embodiment, the grooves **9** have a triangular cross section. Other shapes/cross sections of the grooves are possible in other embodiments. At the distal end of the inner sleeve, one or more passageways **12** spaced radially outwardly from longitudinal axis A, extend through, and distally protrude from, an annular distal wall **13** of the inner sleeve **10**, the purpose of which is to ensure alignment between stabilizer base **38** and the IO device **30** (particularly inner sleeve **10**), by the alignment of passageways **12** of inner sleeve **10** to one of each of notches **41a-c** in stabilizer base **38**. The protrusions **12** are adapted to receive a corresponding probe needle **27** and are aligned with holes **46** in base **40**.

(73) The distal end of inner sleeve **100** also includes one or more protrusions **110** spaced radially outwardly from longitudinal axis A, extend through, and distally protrude from, an annular distal wall **113** of the inner sleeve **100**. The protrusions **110** are aligned with notches **41a** or more (**41b**, **41c** etc.,) provided in base **40**.

(74) As shown in FIG. 5A, the protrusion **110** and passageway **12** may be formed monolithically as one part.

(75) Distal wall **13** has a central opening **6** through which the intraosseous needle assembly is received after the outer sleeve member is mounted about the inner sleeve. A plurality of circumferentially spaced J-shaped attachment elements **2** each of which releasably engageable with a corresponding set of constricting elements **32a** and **32b** (see FIG. 4) protrude distally from the distal edge **1** of outer surface **11**.

(76) FIG. 6 illustrates a cross sectional view of outer sleeve member **15** in one embodiment, showing the inner structure of outer sleeve **20**. FIG. 6A illustrates a cross sectional view of body **4** with motion inducer **28** and needle holder **26** (in outer sleeve **200**), and FIG. 6B illustrates a cross sectional view of body **4** in a cross section rotated 90 degrees relative to FIG. 6A. Outer surface **21** of the outer sleeve **20** is substantially parallel to the longitudinal axis A of the outer sleeve member **15**, which is adapted to be coincident with the longitudinal axis of the inner sleeve. Outer surface **14** of the grip **16** extending from a distal region of outer surface **21** is oblique with respect to

longitudinal axis A.

(77) The inner diameter of outer surface **21** is substantially equal to the outer diameter of outer surface **11** of the inner sleeve in the embodiment shown in FIG. 5.

(78) With continued reference to FIG. 6, an annular intermediate surface **33** is formed with central opening **18** and is substantially perpendicular to longitudinal axis A (shown in FIG. 5). A rigid tube **47** having a window (i.e., aperture) **48** extends distally from intermediate surface **33**. An annular mounting post **51** extends from the inner surface of tube **47** distal to window **48**. In one embodiment, annular mounting post **51** includes external threading **52** at its distal end which protrudes from tube **47**. Two elongated, narrow flexible catches **56** each spaced radially outwardly from tube **47** extend distally from intermediate surface **33**. The distal portion of each catch **56** is configured with rounded teeth **61** at the outer edge thereof to facilitate engagement with grooves **9** of inner sleeve **10** (see FIG. 5), and is configured with a curved, or otherwise oblique, inner edge **62** that extends towards outer surface **21**. The clearance **59** between tube **47** and catches **56** constitutes a spring chamber within which proximal spring **63** (see FIG. 2) is receivable.

(79) While driving member hub **5** is separated from outer sleeve member **15**, Luer-Lock fitting **57** is inserted into the interior of mounting post **51** and engaged with the inner surface of the latter. At the same time, flexible tube **55** connected to Luer-Lock fitting **57** protrudes distally from nut **29**. While Luer-Lock fitting **57** is being inserted into the interior of mounting post **51**, the outer periphery of Luer-Lock fitting **57** forces decoupling element **77** to be displaced radially outwardly momentarily until returning to be displaced radially inwardly under the influence of flexible arm **78** such that the narrowing tip of hook-shaped decoupling element **77** is positioned in pressed engagement with the lip **68** (FIG. 7) of Luer-Lock fitting **57** as shown. Throat portion **7** of driving member hub **5** is then introduced within central opening **18** of outer sleeve member **15** (FIG. 6), so that rod **42** connected to the throat portion will be received within the interior of, and strengthen, tube **55**, until elements of driving member hub **5** are engaged with surfaces **19** and **33** of outer tube member **15**.

(80) FIGS. 7 and 8 illustrate an embodiment with IO needle assembly **35** and a portion of probe needle assembly **25** when assembled and located at one of different possible axial relative positions therebetween. Motion inducer **28** of probe needle assembly **25** includes a discontinuous tubular periphery **73** and with two diametrically opposed rectangular extensions **74** that radially protrude outwardly from terminal edges of periphery **73**. The distal end of each extension **74** includes a wedge-shaped radial expander **76**. A narrow flexible arm **78** is spaced radially inwardly from a corresponding extension **74**, to accommodate positioning of a spring within the radial clearance **79**. The narrow flexible arm **78** is slightly longer than the extension and includes at its proximal end a hook-shaped, snappable decoupling element **77** facing away from periphery **73**. An element **72** interconnects extension **74** and arm **78**. Arm **78** is in abutting relation with rigid tube **47**, and decoupling element **77** is in abutting and pressed relation with a lip **68** extending slightly radially outwardly from the periphery at the proximal end of Luer-Lock fitting **57**.

(81) Rod **42** of IO needle assembly **35** is fully inserted/contained within the lumen of flexible tube **55**. The proximal end of the tube being integrally formed with, or connected by, a leak-free connection. In various embodiments, such connections may include a threaded connection, adhesion or overmolding to Luer-Lock fitting **57**. Rod **42** extends through the interior of Luer-Lock fitting **57** so that its proximal end is receivable in the axial bore formed in the driving member hub throat portion. The blunt distal end **43** of rod **42** is in contact with the solid distal end of bone portal **60**.

(82) In the embodiment shown in FIGS. 9 and 9A, bone portal **60** is a sharp stainless-steel member having a varying diameter. A proximal portion **83** of bone portal **60** is hollow and a distal portion **84** thereof is solid and uncompromised. The distal portion **84** of bone portal **60** terminates with a high strength pointed distal end **86** that is suitable to penetrate the bone cortex and to access the bone marrow cavity. Solid distal portion **84** of bone portal **60** may be configured with three sharp-

angled facets **97a-c**. Bone portal **60** may be connected to flexible tube **55** by a physical connection as will be described below, or may be integral as one unit in an over molding process.

(83) Sharp barbs **91** protrude from the periphery of proximal portion **83** to facilitate radial connection with flexible tube **55**. The barbs **91** preferably protrude obliquely and distally from proximal portion **83** to resist detachment from flexible tube **55** during withdrawal of the bone portal **60** from the bone cortex. The inner diameter of flexible tube **55** may be greater than or equal to the outer diameter of barbs **91**, allowing the flexible tube **55** to be positioned over the barbs **91** so that the barbs will be able to bite into the wall of the flexible tube when the latter is squeezed, such as by means of crimp ring **58**. In various embodiments, crimp ring **58** is formed from a metal material, for example, stainless steel.

(84) A radial abutment **93** extends radially outwardly from the outer wall of proximal portion **83**, at a region distal to the barbs **91**. Crimp ring **58**, when fixed in encircling and pressing relation with respect to the outer surface of flexible tube **55**, is secured by an annular distal surface thereof with the distal surface of radial abutment **93**. Radial abutment **93** thus fulfills several functions. Firstly, it distally supports and abuts flexible tube **55** both when being attached to barbs **91** and during a transfusion operation. Abutment **93** constitutes an anchoring point for the proximally positioned flexible tube **55**, so that forces generated during penetration into the bone cortex will be suitably distributed to pointed distal end **86** to prevent collapse of flexible tube **55**. When flexible tube **55** is attached, its outer wall is substantially aligned with the outer edge of abutment **93**. Secondly, radial abutment **93** constitutes means by which crimp ring **58** is secured. Thirdly, the abutment is able to contact the outer surface of the bone cortex during a penetration operation, serving as stopping means to prevent overpenetration of the bone portal into the bone when excessive force is applied to the driving member hub.

(85) It will be appreciated that a radial abutment **93** or any other type of shoulder can be provided with other elements of the IO catheter **92**, such as the driving rod member **42**, the flexible tube **55**, and the crimp ring **58**.

(86) One or more bores **99** are provided in bone portal **60** in order to facilitate fluid exchange with the bone marrow cavity. Bores **99** are in fluid communication with internal lumen **96** of proximal portion **83** and with internal lumen **98** of flexible tube **55**, through which infusion liquids are flowable. In various embodiments, each bore **99** may be elliptically shaped and extend radially outwardly from lumen **96** at an interface between proximal portion **83** and distal portion **84**. The elliptical shape prevents bone chips from entering the bores and lumen **96** during bone penetration. When two bores **99** are employed, they may be positioned diametrically opposite from each other. The liquid flow through the one or more elliptic bores **99** may be equal to the flow that is achievable through a G15 hollow needle or other selected needle gauges.

(87) The stages (or steps) of a penetration operation according to various embodiments will now be described.

(88) The IO device **30** is removed from the interior of package **105** shown in FIG. **17** after cover **102** is separated from package body **103**, and ready for use. This may include removal of safety latch **45** and release liner **66** from the skin-facing side of fixation sticker **65** to expose the adhesive.

(89) FIG. **10** illustrates a pre-penetration cross-sectional view of one embodiment of the IO device **30**, along a plane orthogonal to (i.e., angularly separated by 90 degrees from) the plane along which the cross-sectional view of FIG. **1A** is taken, showing the IO device **30** positioned adjacent a patient's skin/soft tissue overlying his or her bone.

(90) At the initial stage, the rounded teeth **61** of each catch **56** are spaced proximally from grooves **9** of inner sleeve **10**, and the distal end of bone portal **60** is distally spaced from the distal end of probe needles **27** by distance X. Distance X is greater than the average bone cortex thickness and less than the average internal diameter of a marrow cavity, and may be any distance between 1.5 mm and 5 mm, e.g., 3 mm. Each radial expander **76** of motion inducer **28** thereof is in contact with the inner edge **62** (see FIG. **6**) of a corresponding catch **56**, and each decoupling element **77**

protrudes radially inwardly through window **48** of tube **47**.

(91) FIGS. **11** and **11A** illustrate IO device **30** after the safety latch **45** has been disengaged and a driving force (i.e., manual pressure from a user) has been applied to driving member hub **5** that causes outer sleeve member **15** to be distally displaced with respect to inner sleeve **10**, **100** (i.e., in a collapsing/telescoping motion). The probe needles **27** and IO needle assembly **35** are distally displaced in unison, protruding through stabilizer **40** to be exposed and then penetrate through the patient's soft subcutaneous tissue, and the decoupling elements **77** remain in pressed engagement with the lip of Luer-Lock fitting **57**. Due to the distal displacement of outer sleeve member **15** relative to inner sleeve **10**, teeth **61** of the catch slide along the inner surface **8** of inner sleeve **10** which is formed with grooves **9**.

(92) At the second stage, the driving force/manual pressure for initiating distal displacement of outer sleeve member **15** is transmitted to needle holder **26**, and the latter in turn transmits the driving force to distal spring **64**, causing it to be compressed. If the thickness of the soft subcutaneous tissue is less than distance X, the bone probe of IO needle assembly **35** is likely to start penetrating the bone cortex. Since the penetration depth limiting mechanism has not yet been activated, outer sleeve member **15** will return to its position at the initial stage upon cessation of the driving force and upon the subsequent expansion of distal spring **64**, which proximally propels needle holder **26** and the entire outer sleeve member **15** that is connected to the needle holder.

(93) In the third stage illustrated in FIGS. **12** and **12A**, IO device **30** is shown after outer sleeve member **15** has been additionally displaced distally, until the distal end of probe needles **27** contacts the bone cortex. As a result of the reactive force applied by the bone cortex onto the probe needles **27**, the latter become immobilized and are therefore prevented from undergoing additional distal displacement. A distal end of each of the probe needles **27** may be configured with a radial protrusion greater than the main probe needle structure, in order to intensify the reactive force applied by the bone cortex onto the radial protrusion of a probe needle.

(94) Disengagement of probe needles **27** from the bone cortex is prevented by means of proximal spring **63**, such that the spring force applied thereby (which may be designed to correspond to the bone's reactive force) presses probe needles **27** onto, but not through, the bone cortex. The decoupling elements **77** remain in pressed engagement with the lip of Luer-Lock fitting **57** and the relative position of a radial expander **76** and the corresponding catch inner edge **62** remains unchanged. Although the distal tip of IO needle assembly **35** has penetrated the bone cortex and may have even penetrated the marrow cavity during this third stage and remain separated by the distal end of probe needles **27** by distance X, the distal tip of IO needle assembly **35** is able to be withdrawn upon cessation of the driving force since the penetration depth limiting mechanism has not yet been activated.

(95) FIGS. **13** and **13A-B** illustrate wherein IO device **30** in the fourth stage, after the manual driving force continues to be applied onto driving member hub **5** and/or outer sleeve member **15** following the immobilization of probe needles **27** during the third stage. The driving force is transmitted through the intermediate surface **33** of outer sleeve member **15** to proximal spring **63**, causing it to become compressed upon contact with the immobilized proximal end **79** of each probe needle **27**. Proximal spring **63** is biased with a spring force, for example approximately 1.5 kgf, which enables movement of probe needles **27** through the subcutaneous tissue, but proximal spring **63** will compress in response to application of the reactive force by the bone cortex once the probe needles **27** become immobilized.

(96) As a result of the compression of proximal spring **63**, driving member hub **5** and IO needle assembly **35** connected thereto are distally displaced relative to probe needle assembly **25**. IO needle assembly **35** is consequently caused to penetrate the bone cortex, if it was not already penetrated in one of the previous steps. During the distal displacement of driving member hub **5**, throat portion **7** transmits the driving force to lip **68** of Luer-Lock fitting **57**. The distally directly force applied by lip **68** onto the narrowing tip of each hook-shaped decoupling element **77** causes

the decoupling elements to undergo a jerking motion whereby they are displaced radially outwardly and released from lip **68**, and their position and the position of the entire probe needle assembly **25** relative to the distally displaced outer sleeve member **15** is changed. Consequently, decoupling elements **77** are caused to be proximally spaced by a position slightly beyond lip **68** but distal to notch **82**, which is radially recessed from the throat portion **7** of driving member hub **5** at a region near the distal edge of the throat portion.

(97) The combination of driving member hub **5**, motion inducer **28**, Luer-Lock fitting **57**, proximal spring **63**, and decoupling elements **77** as described above may be considered as a release mechanism.

(98) As the driving force continues to be applied during the fifth stage shown in FIGS. **14** and **14A-B**, catch **56** is urged additionally radially outwardly by the radial expander **76** while its teeth **61** are received in the corresponding grooves of inner sleeve **10**, causing outer sleeve member **15** to be mechanically locked to inner sleeve **10** and preventing any additional distal displacement of the outer sleeve member. The distal end of IO needle assembly **35** is thus deployed to the predetermined and maximum depth within the bone cortex which is separated from the distal end of probe needles **27** by distance Y which is greater than distance X of FIG. **11**, being able to access the bone marrow cavity (see FIG. **15A**). Distance Y is the sum of distance X and the incremental axial displacement of the outer sleeve member relative to the inner sleeve during the fourth and fifth stages, e.g., 3 mm.

(99) The combination of driving member hub **5**, inner sleeve grooves **9**, catch **56**, proximal spring **63**, and radial expander **76** as described above may be considered as a penetration depth limiting mechanism. At the same time, the distal displacement of outer sleeve member **15** relative to motion inducer **28** causes the decoupling elements **77** to be received within each corresponding notch **82**, resulting in an audible and tactile click that is indicative to the user that IO needle assembly **35** has been deployed to the predetermined depth.

(100) In addition to the audible feedback sensed by the user upon achieving penetration to the predetermined depth, the user also advantageously receives terminal feedback. That is, the user is made aware that no further penetration into the bone is possible after the predetermined depth has been achieved by being unable to move outer sleeve member **15**. Instead of the feeling sensed during stages 2-5 that outer sleeve member **15** was being distally displaced in response to application of the driving force, the user senses a lack of outer sleeve distal movement, or even a lack of any outer sleeve movement, despite the application of a driving force onto driving member hub **5**. This lack of outer sleeve distal movement constitutes terminal feedback that provides a more pronounced and longer-duration feedback than the one-time audible feedback provided by the decoupling element **77** since it provides an indication of penetration to the predetermined depth each time a user attempts to cause additional penetration into the bone cortex.

(101) In one embodiment, following activation of the penetration depth limiting mechanism, inner sleeve **10**, **100** and outer sleeve member **15** are non-detachably coupled together. In one embodiment, the distal and proximal movement between inner sleeve and outer sleeve member is prevented so that spring **64** remains compressed and the probe needles **27** are exposed. In another embodiment, proximal movement between inner sleeve **10**, **100** and outer sleeve member **15** is possible, as discussed herein. Following actuation of the release mechanism, the outer sleeve member **15** is separated from the IO catheter **92**, and therefore the coupled inner sleeve **10**, **100** and outer sleeve member **15** constitute a releasable component assembly that also includes the coupled motion inducer **28**. Thus, in the sixth stage (see FIG. **15B**), the releasable component assembly is pulled and proximally displaced away from the IO catheter **92**, so that the transfusion-facilitating components remain penetrated within the bone cortex and in communication with the bone marrow cavity while being accessible to a connector interface that is coupled to a transfusion-facilitating component during a transfusion operation.

(102) Alternatively, the penetration depth-limiting mechanism may be configured such that, when

outer sleeve member **15** is mechanically locked to inner sleeve **10**, distal movement of the outer sleeve member relative to the inner sleeve **10**, **100** is prevented while proximal movement of the outer sleeve member relative to the inner sleeve is made possible, for example, in conjunction with the distal spring **63**. Dedicated shaping of the radial expander teeth and of the inner sleeve grooves facilitate the outer sleeve member **15** and the inner sleeve **10**, **100** to be non-detachably coupled together in a fashion that is sensitive to the direction of axial displacement. In this embodiment the inner sleeve covers the probe needles **27** and rod **42**, to prevent unintentional pricking of the patient and/or health practitioner while detaching the IO device.

(103) FIGS. **15B** and **16** illustrates a seventh stage whereby IO catheter **92** remains penetrated within the bone cortex. FIG. **16** illustrates how the flexible tube **55** is plastically reshaped so as to be supported by a curved supporting surface **49** provided in one of the notches **41a-c** formed in base **38** of stabilizer **40**. This prevents formation of a crease or kink in the flexible tube **55**, which may restrict flow of infusion liquids therethrough. The reshaped flexible tube **55** is secured in place by constricting elements **32a** and **32b** thereon. Although flexible tube **55** is shown to be supported and secured in one of the notches formed in based **38**, it will be appreciated that the flexible tube can likewise be supported and secured in any of the other notches, depending on the given position and degree of comfort or convenience of the user.

(104) By fixating Luer-Lock fitting **57** at the proximal end of flexible tube **55** at such a low profile close to base **38** and also adjacent to the skin of the patient, the risk of unintentional withdrawal of IO catheter **92** from the bone due to patient or health practitioner movement is essentially prevented. Also, the three notches **41a-c** provided in base **38** advantageously allow the health practitioner to secure IO catheter **92** in one of three different directions depending on other treatment related issues, in order to initiate a penetration or a transfusion operation, thereby enhancing the convenience and ease of use.

(105) At the end of the transfusion insertion operation, the used IO device with the probe needles **27** may be disposed of in the original IO device package **105** shown in FIG. **17**, thereby transforming the package into a container for biological waste. Additionally, if needed, the package **105** can serve as a container for biological waste for IO catheter **92** at the end of the transfusion operation.

(106) In embodiments where the proximal movement of the outer sleeve member **15** over the inner sleeve **10**, **100** is possible, the safety latch **45** may be coupled again between outer sleeve **20** and inner sleeve **10**, **100** at the end of the insertion procedure, thereby preventing unintentional pricking of the patient and/or health practitioner by the probe needles **27**. The used IO device **92** may then be safely disposed in a biological waste bin (i.e., a sharps container).

(107) It will be appreciated that the base **38** and stabilizer **40** may be configured in other ways as well insofar as the IO catheter **92** and probe needles **27** are afforded a passageway to subcutaneous tissue and to the adjacent bone cortex. Likewise, the base **38** and stabilizer **40** may be dispensed with when the releasable component assembly is suitably held during a penetration operation.

(108) In another embodiment, a flexible tube is not employed, but rather an inflexible annular IO catheter is used. The IO catheter is connected to or integrally formed with the Luer-Lock fitting and with the bone portal. The driving member, which is connected to the driving member hub and in driving engagement with the Luer-Lock fitting, is inserted into the lumen of the IO catheter, and the penetration operation can be performed as described above.

(109) In another embodiment, a flexible tube and a separate driving member connected to the driving member hub are not employed. In this embodiment, the IO catheter comprises a rigid rod constituting the driving member that is integrally formed with the Luer-Lock fitting and with the bone portal and formed within an axial passageway through which liquid is flowable. The driving member hub is in driving engagement with the Luer-Lock fitting and the penetration operation can be performed as described above.

(110) In other embodiments, the driving member hub is fixedly and detachably connected to outer

sleeve member **15** by one of various means such as a Bayonet connector, a threaded connection, and a snap connection. Such a detachable connection will allow the IO catheter to be reassembled onto the driving member, for example, when used for training purposes.

(111) FIG. **18** illustrates an embodiment of a driving member hub **145** which is configured with a secure and speedy to couple Bayonet-shaped connector. Driving member hub **145** includes a mount **143** for a plurality of outwardly protruding pins **149** of the Bayonet-shaped connector that is fixed to the distal side of a domed proximal surface **141** of the hub. The domed proximal surface **141** may have a raised/embossed/grooved pattern to improve a user's grip on/manual engagement thereof (also see the similar pattern on the surface of hub **5** in FIG. **1**). Each pin **149** is configured to be inserted into a corresponding groove formed in the annular wall **44** of outer sleeve member **15** which is proximal to intermediate surface **33** (see FIG. **6**). Each of the grooves, for example, grooves **87** and **88**, may be differently shaped. After a pin **149** is received in each corresponding groove, driving member hub **145** is rotated in a first rotational direction to lock the pins **149** in place. Pins **149** may be loosened and then detached from the grooves upon rotating driving member hub **145** in a second rotational direction which is opposite to the first rotational direction.

(112) In this embodiment, tube **147** is formed with windows (i.e., apertures) **148** within each corresponding decoupling element and is received during the fifth stage of the penetration operation to produce an audible and tactile click when the IO needle assembly has been penetrated to the predetermined depth in the patient's bone cortex. The tube **147** extends distally from throat portion **142**, which is attached to the distal surface of mount **143**. Throat portion **142** has a slightly larger diameter than tube **147**. The distal end of tube **147** is formed with a notch **152** through which the lip of the Luer fitting radially extends when the distal side of the lip is secured by the decoupling elements during the first, second and third stages of the penetration operation.

(113) While some embodiments of the invention have been described by way of illustration, it will be apparent that the invention can be carried out with many modifications, variations and adaptations, and with the use of numerous equivalents or alternative solutions that are within the scope of persons skilled in the art, without exceeding the scope of the claims. Although embodiments have been disclosed, the invention is not limited thereby.

(114) All of the features disclosed in this specification may be combined in any combination. Each feature disclosed in this specification may be replaced by an alternative feature serving the same, equivalent, or similar purpose. Thus, unless expressly stated otherwise, each feature disclosed is only an example of a generic series of equivalent or similar features.

(115) From the above description, one skilled in the art can easily ascertain the essential characteristics of the present invention, and without departing from the spirit and scope thereof, can make various changes and modifications of the invention to adapt it to various usages and conditions. Thus, other examples are also within the claims.

(116) In general, any combination of disclosed features, components and methods described herein is possible. Steps of a method can be performed in any order that is physically possible.

(117) All cited patents, patent publications and non-patent references are incorporated by reference herein in their entireties.

Claims

1. A manual intraosseous device for introducing a bone portal to a predetermined depth relative to a surface of a target bone, comprising: an intraosseous needle assembly including an intraosseous catheter configured to facilitate transfusion of fluids and aspiration of bone marrow and having a bone portal adapted to penetrate into subcutaneous tissue adjacent the target bone and the cortex of the target bone and a flexible tube configured to convey liquids to and from the bone marrow of the target bone, and a driving member releasably attached to the intraosseous catheter and by which a manual axial force is transmittable to the intraosseous catheter to initiate axial penetration into the

subcutaneous tissue and cortex of the target bone; an inner tubular sleeve arranged concentric to the intraosseous catheter; an outer sleeve member including an outer sleeve arranged concentric to the inner sleeve and being axially displaceable relative to an outer surface of the inner sleeve and the outer sleeve member operably connected to the driving member, and wherein the driving member engages the outer sleeve member; a probe needle assembly engaged with the outer sleeve member and at least partially received within an interior of the inner sleeve, and including at least one bone cortex-contacting probe needle; a proximal spring positioned within the outer sleeve member, operably connected to the probe needle assembly, and configured to limit the depth that the bone portal penetrates the target bone; and a distal spring positioned within the inner sleeve, operably connected to the probe needle assembly, and configured to move the outer sleeve member to a starting position if the bone portal penetrates the target bone cortex to the predetermined depth, to prevent any additional distal displacement of the bone portal; and a stabilizer including a base configured to detachably couple with a distal end of the inner sleeve and having at least one notch sized and shaped to receive at least a portion of the flexible tube therein.

2. The intraosseous device of claim 1, further comprising a body disposed within the outer sleeve and in a concentric relation with inner sleeve.

3. The intraosseous device of claim 2, wherein the body includes an annular mounting post, and further comprising a motion inducer for securing the at least one probe needle, the motion inducer being insertable through the annular mounting post.

4. The intraosseous device of claim 3, wherein the motion inducer includes a discontinuous tubular periphery and two diametrically opposed rectangular extensions that radially protrude outwardly from terminal edges of periphery, the distal end of each extension including a wedge-shaped radial expander.

5. The intraosseous device of claim 1, wherein an inner surface of the outer sleeve includes at least one axial slot formed therein, and wherein the outer surface of inner sleeve includes at least one protrusion protruding radially outwardly therefrom and configured to insertably engage the at least one axial slot, whereby the engagement of the at least one protrusion within the at least one axial slot maintains the inner sleeve and outer sleeve in a concentric relation during axial displacement.

6. The intraosseous device of claim 1, wherein the outer sleeve includes a stopper protruding inwardly from the inner surface of outer sleeve and configured to limit the proximal displacement of outer sleeve relative to inner sleeve.

7. The intraosseous device of claim 1, wherein the distal displacement of the outer sleeve member relative to the motion inducer causes an audible and tactile click that is indicative to the user that the intraosseous needle assembly has been deployed to the predetermined depth.

8. The intraosseous device of claim 1, further comprising a removeable safety latch disposed between the outer sleeve and the inner sleeve and configured to prevent displacement of the driving member due to an unintentional operation of the intraosseous device.

9. The intraosseous device of claim 1, wherein the stabilizer has an aperture sized and shaped to receive at least a portion of the intraosseous needle assembly.

10. The intraosseous device of claim 9, wherein the stabilizer includes a plurality of notches formed in the base and configured to secure the flexible tube therein.

11. The intraosseous device of claim 10, wherein the stabilizer includes a planar supporting surface having a plurality of terminal edges that are spaced from and facing a corresponding notch, and a plurality of guidable peripheral edges extending between two adjacent terminal edges.

12. The intraosseous device of claim 11, wherein the guidable edges are shaped so as to be positionable close to a prominent anatomical feature and are located at a predefined distance from a center of base to assist in properly positioning stabilizer relative to the target bone to be penetrated.

13. A manual intraosseous device for introducing a bone portal to a predetermined depth relative to a surface of a target bone, comprising: an intraosseous catheter configured with a driving member and a bone portal that is penetrable into a cortex of the target bone in conjunction with the driving

member, the driving member configured to support the intraosseous catheter; an inner tubular sleeve; an outer tubular sleeve concentric to, and axially displaceable relative to, the inner sleeve; wherein the driving member is connected to the outer sleeve, by which a manual axial force applied to the driving member is transmittable to the intraosseous catheter to initiate corresponding axial penetration into subcutaneous tissue associated with the target bone and into the cortex of the target bone; and a penetration depth limiting mechanism to prevent any additional distal displacement of the bone portal when the bone portal is penetrated within the bone cortex; and a stabilizer including a base configured to detachably couple with a distal end of the inner sleeve and defining at least one notch configured to secure the flexible tube therein.

14. The intraosseous device of claim 13, further comprising an outer sleeve member that includes the outer sleeve and a grip engaging an outer surface of outer sleeve, the outer sleeve member operably connected to the driving member.

15. The intraosseous device of claim 14, wherein the driving member further includes a rod, and a driving member hub configured to engage the outer sleeve member and includes a throat portion that is configured to receive a proximal end of the rod therein.

16. The intraosseous device of claim 14, wherein the penetration depth limiting mechanism includes a proximal spring positioned within the outer sleeve member, configured to limit the depth that bone portal penetrates the target bone; and the penetration depth limiting mechanism includes a distal spring positioned within the inner sleeve, operably connected to the probe needle assembly, and configured to move the outer sleeve member to a starting position if the bone portal penetrates the target bone cortex to the predetermined depth, to prevent any additional distal displacement of the bone portal.

17. The intraosseous device of claim 13, wherein the intraosseous catheter is further configured with an internal lumen through which an infusion liquid or a bone marrow aspirate is flowable, and at least one port by which the internal lumen is positionable in fluid communication with a bone marrow cavity of the target bone, and is connectable with a connector interface that induces flow of the infusion liquid or the bone marrow aspirate when the driving member hub is released from the intraosseous catheter.

18. The intraosseous device of claim 13, further comprising a body disposed within the outer sleeve and in a concentric relation with inner sleeve.

19. The intraosseous device of claim 18, wherein the body includes an annular mounting post, and a motion inducer insertable through the annular mounting post.

20. The intraosseous device of claim 13, wherein the stabilizer has an aperture sized and shaped to receive at least a portion of the intraosseous catheter.
